ps1-av-capture, one channel. TVP7002 digitises the PS1 analog RGB, an FX3 bridges it to USB 3 (UVC firmware, AN75779 pattern), a PCM2903C presents the line audio as a USB Audio Class device on its own port. Design notes: docs/design.md.

Input

File: input.kicad_sch

TVP7002

File: tvp7002.kicad_sch

FX3

File: fx3.kicad_sch

Power

File: power.kicad_sch

Audio

File: audio.kicad_sch

rixnobis

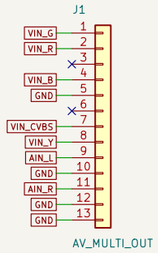
Sheet: /

File: ps1-av-capture.kicad_sch

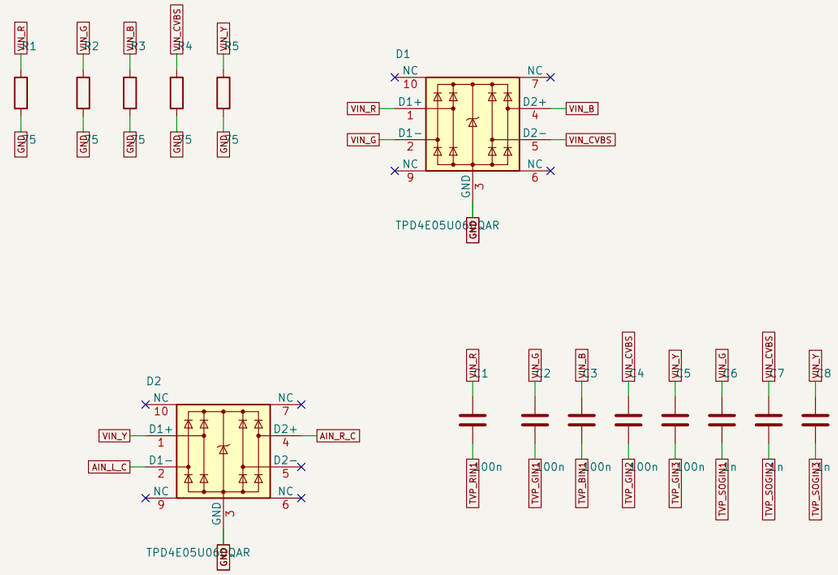
Title: ps1-av-capture: PS1 RGB + audio to USB

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/6

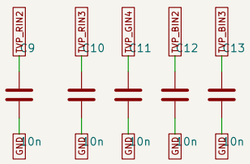
PS1 AV multi-out (psx-spx pinout). Pin 3 (+5V) is not taken: the board is powered from USB. Pin 6 (S-Video C) is not used.



75 ohm termination at the connector, then ESD, then AC coupling (TVP7002: 0.1 uF; SOG inputs: 1 nF). CVBS and Y also reach spare ADC inputs, so luma can be digitised.



Unused TVP7002 analog inputs: 10 nF to GND (SLES206C Table 1).



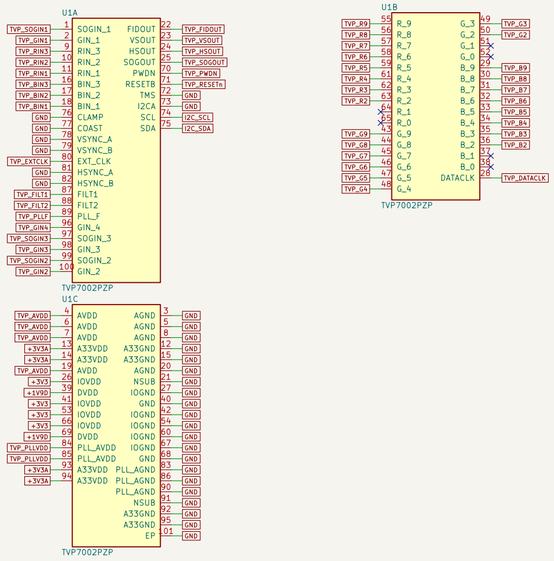
rixnobis

Sheet: /Input/
File: input.kicad_sch

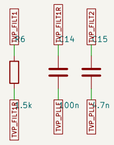
Title: AV multi-out input, termination, ESD

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/6

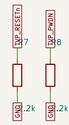
TVP7002. Sync comes from SOGIN (composite or luma; the PS1 has no sync pin), so the digital HSYNC/VSYNC inputs are grounded as unused. I2C address B8h (I2CA low).



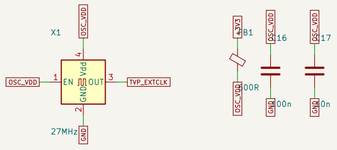
H-PLL loop filter (SLES206C Figure 4): 1.5k + 100n FILT1–PLL_F, 4.7n FILT2–PLL_F.



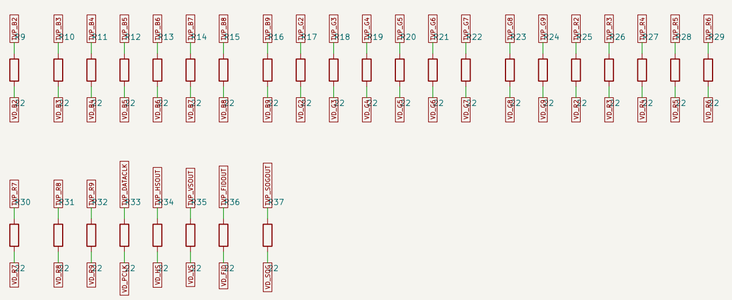
Control. RESETB is held low by 2.2k until the FX3 firmware releases it, so the TVP7002 stays in reset while the FX3 boots from the shared I2C bus (SLES206C: no I2C activity before reset completes). PWDN 2.2k to GND (Figure 8).



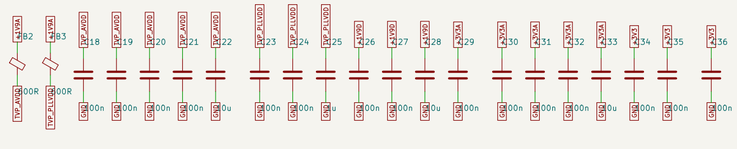
27 MHz reference on EXT_CLK for mode detection (SLES206C recommends 27 MHz; the internal reference is about 6.5 MHz and imprecise).

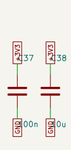


Output bus: 22R series at the source (FX3 AN70707 checklist item 9). Top 8 bits of each channel. 4:4:4 uses DQ[23:0]; 20-bit 4:2:2 uses DQ[15:0] (Y on G, CbCr on B).



Supply decoupling. SLES206C gives no values; this follows the OSSC: ferrites on AVDD and PLL_AVDD, 100n per pin, bulk per rail.





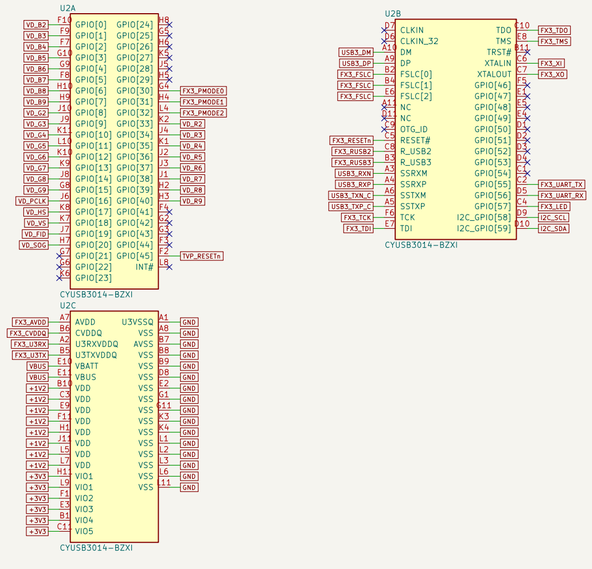
rixnobis

Sheet: /TVP7002/
File: tvp7002.kicad_sch

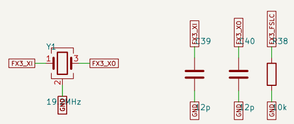
Title: TVP7002 RGB digitiser

Size: A2 | Date: | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 3/6

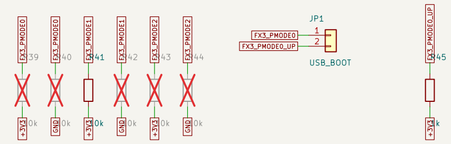
CYUSB3014. GPIF II takes the TVP7002 bus: DQ[7:0]=B[9:2], DQ[15:8]=G[9:2] (VIO1), DQ[23:16]=R[9:2] (VIO2). PCLK=DATACLK, CTL[0..3]=HS/VS/FID/SOG. All VIO at 3.3 V.



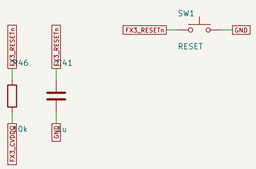
19.2 MHz crystal, FSLC[2:0]=000 through one 10k (AN70707 Figure 9). No series resistor on the crystal (AN70707). Load caps sized for a 12 pF crystal with ~5 pF stray.



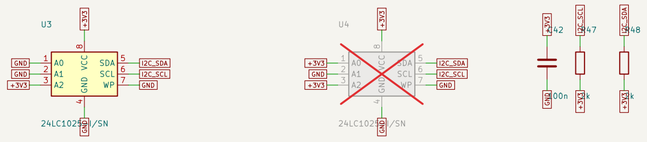
Boot: PMODE F1F = I2C EEPROM, USB fallback (AN70707 Table 9). 10k up/down footprints on every PMODE pin, only PMODE1 up fitted. Jumper JP1 pulls PMODE0 high = F11, USB boot only.



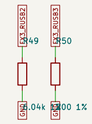
Reset: RC holds RESET# low > 1 ms after power (tRPW 1 ms, tRR 5 ms with a crystal); button and the JTAG header can also pull it.



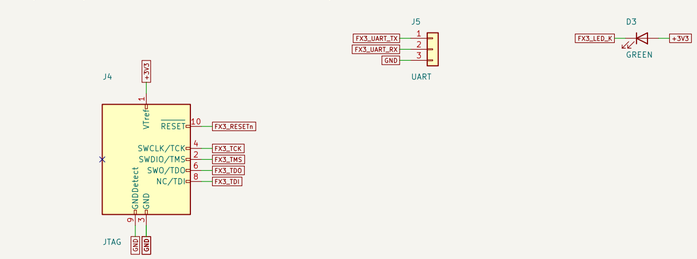
I2C: boot EEPROM and TVP7002 on one bus, 2k pull-ups to VIO5 (AN70707). 24LC1025 A2 must be tied high; A1:A0 = 00.



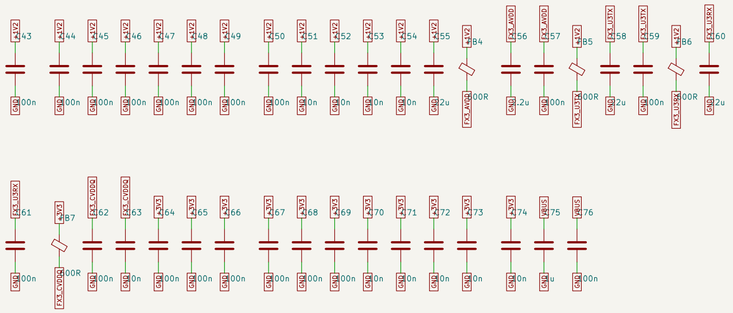
USB reference resistors (AN70707): R_USB2 6.04k 1%, R_USB3 200R 1%.



Debug: JTAG (no external pulls; FX3 has internal ones). UART on GPIO55/56, LED on GPIO57.



Decoupling per AN70707 Table 3. AVDD, U3TXVDDQ, U3RXVDDQ and CVDDQ each behind a ferrite (AN70707 Figure 1, Appendix B); 22 uF on each USB 3 supply (datasheet).

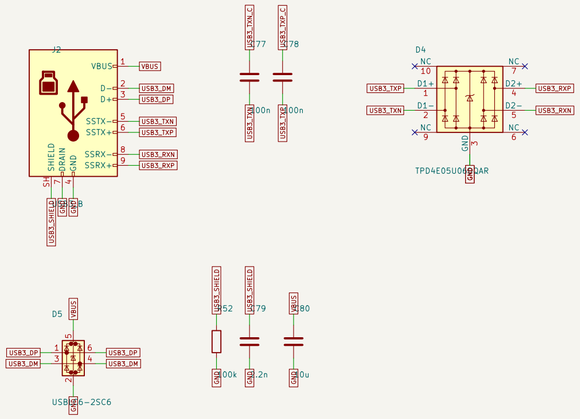


rixnobis
Sheet: /FX3/
File: fx3.kicad_sch

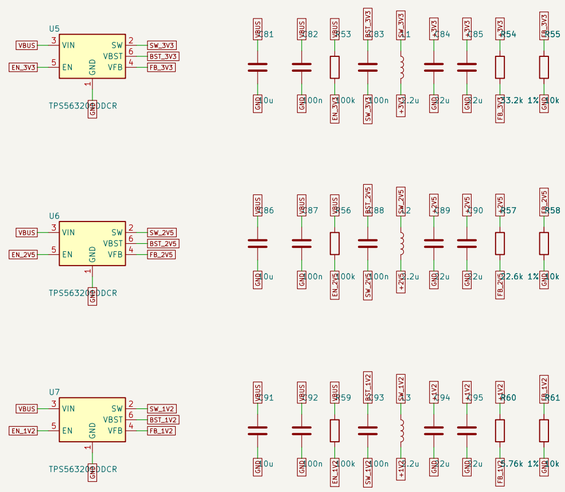
Title: FX3 USB 3 bridge

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.7		Id: 4/6

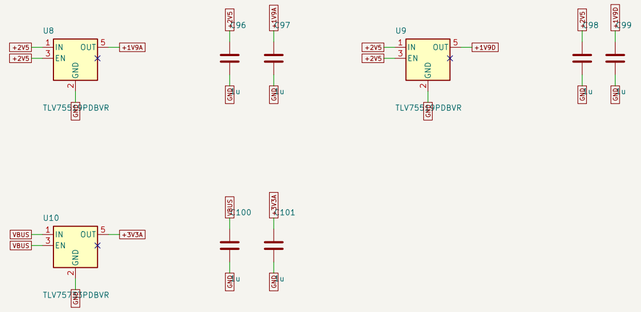
USB 3 Standard-B (FX3). SS TX AC-coupled with 100 nF at the connector (AN70707 p.22, p.40). ESD at the connector. Shield to GND through 100k || 2.2 nF (AN70707 Figure 35).



Three TPS563201 bucks from VBUS (SLVSD90B Table 7-2: R2 = 10k, L = 2.2 uH, 2 x 22 uF out). 3.3 V digital; 2.5 V pre-regulator for the TVP7002 1.9 V LDOs (OSSC pattern); 1.2 V FX3 core. EN has an internal pull-down, so it is tied up through 100k.



LDOs: 2.5 V -> 1.9 V for TVP7002 AVDD/PLL and DVDD (two TLV75519, 500 mA); VBUS -> 3.3 V for TVP7002 A33VDD alone (TLV75733, 1 A), kept off the buck rail.



Power LED, mounting holes.

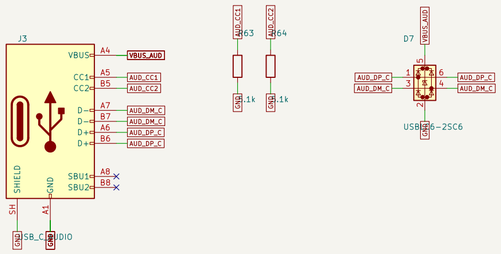
Power flags (ERC): +1V2, +2V5, +3V3, FX3_AVDD, FX3_CVDDQ, FX3_U3RX, FX3_U3TX, OSC_VDD, TVP_AVDD, TVP_PLLVDD, VBUS_AUD

rixnobis
Sheet: /Power/
File: power.kicad_sch

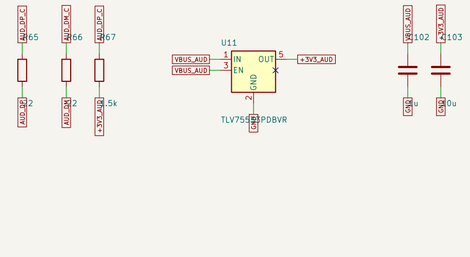
Title: USB 3 connector and power

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.7		Id: 5/6

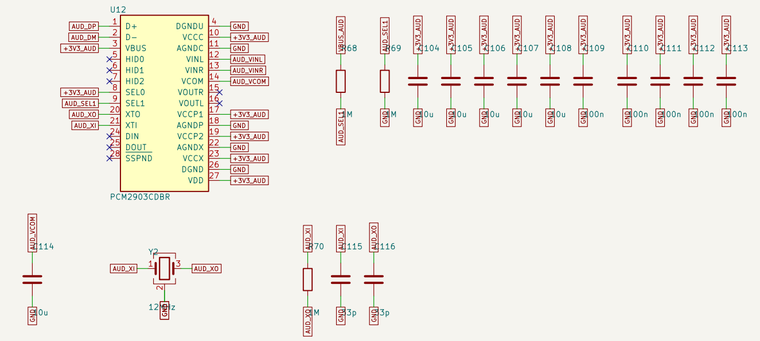
PCM2903C on its own USB 2.0 Type-C port. TI documents the PCM2903C self-powered only (SBFS038A Figure 41), so a 3.3 V LDO makes its supply from this port's VBUS.



D+/D−: 22R series; 1.5k D+ pull-up on the connector side (Figure 41). TI gates the pull-up with an AND of VDD and VBUS; here VDD only exists when VBUS does, so it goes to +3V3_AUD.



PCM2903C wiring per Figure 41: VBUS pin and SEL0 to VDD; SEL1 to the port VBUS with 1M to GND; 10 uF on every supply pin and VCOM; 12 MHz crystal with 1M across it. DAC, S/PDIF and HID pins unused.



Line input: 2.2 uF coupling, then 10k into the 30k ADC input (x0.75, so 2.6 Vpp maps to the 1.98 Vpp full scale) with 330 pF to GND (anti-alias pole ~48 kHz).

rixnobis

Sheet: /Audio/
File: audio.kicad_sch

Title: PCM2903C USB audio

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.7		Id: 6/6